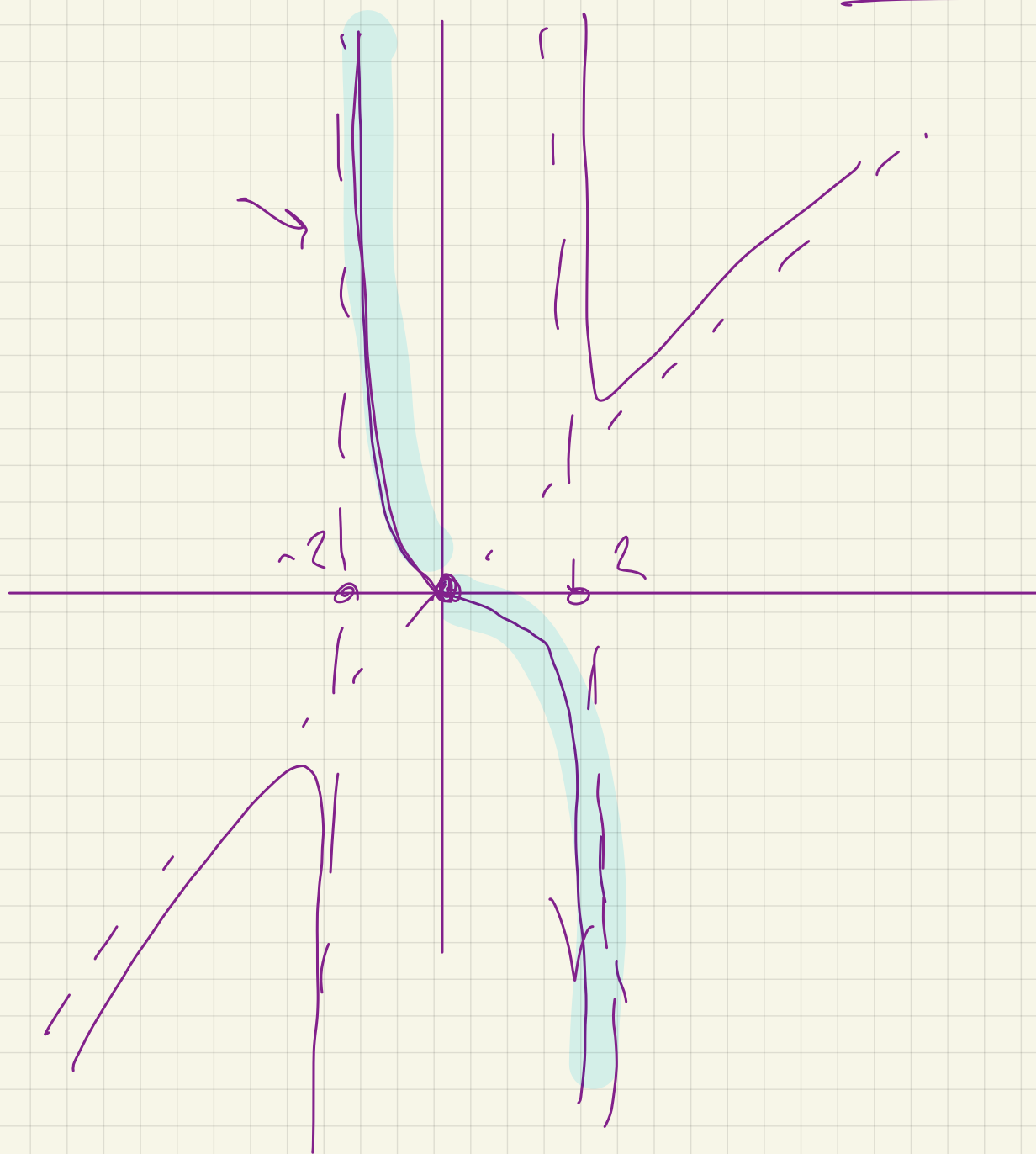




$$f(x) = \frac{x^3 + x}{x^2 - 4} \quad \nearrow \infty$$

$$x^2 \neq 4$$

$$x \neq \pm 2$$



$$\ln \frac{x}{y} \Rightarrow \ln \frac{kx}{ky} = \ln \frac{x}{y} \Rightarrow$$

$$\underbrace{k^0 \ln \frac{x}{y}}_1$$

логарифм с основанием 0

φ-не однород

$$f(x, y) = k^n f(kx; ky)$$

$$\searrow e^{\frac{kx}{ky}} \Rightarrow e^{\frac{x}{y}}$$

$$f(x, y) \rightarrow k^0 \cdot f(kx; ky) = f(x, y)$$

$$f(x, y) = x^2 y + y^2 z + z^2 x$$

$$f(kx; ky) = k^2 x^2 ky + k^2 y^2 kz + k^2 z^2 kx$$

$$= k^3 x^2 y + k^3 y^2 z + k^3 z^2 x =$$

$$= k^3 (x^2 y + y^2 z + z^2 x) =$$

$$k^3 f(x; y)$$

3 morph.

$$f(x; y) = k^n f(kx; ky) \Rightarrow \text{homog}$$

$$f(x; y) = x^2 \sin \frac{x}{y}$$

$$\underline{f(kx; ky) = k^2 x^2 \sin \left(\frac{kx}{ky} \right) = k^2 x^2 \sin \frac{x}{y} = k^2 f(x; y)}$$

$$f(x; y) = x^a \cdot y^{1-a}$$

$$f(kx; ky) = k^a x^a \cdot k^{1-a} y^{1-a} =$$

$$k^a \cdot k^{1-a} x^a y^{1-a} = k^{a+1-a} x^a y^{1-a}$$

$$= k^1 x^a y^{1-a} = k x^a y^{1-a}$$

$$f(x; y) = \underbrace{x + y + 1}$$

$$f(kx; ky) = kx + ky + 1 = k(x + y) + 1 = \frac{k(x + y) + 1}{k} = k f(x; y)$$

ult.

k positive токен

k positive !

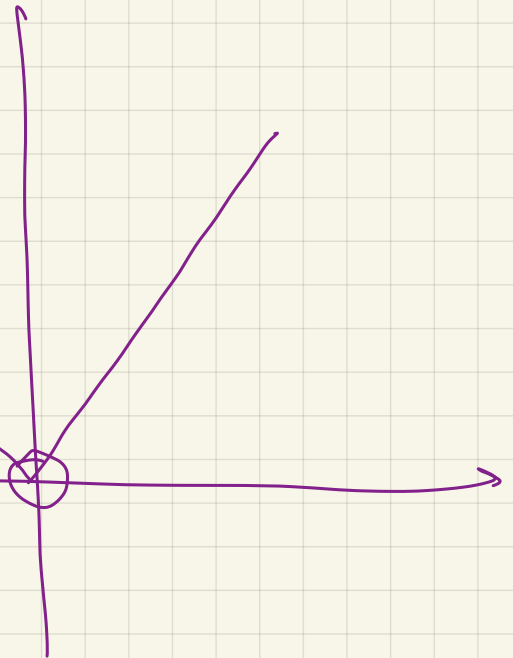
$$x_0 \in \mathbb{D}$$

$$\underline{f'(x_0) = 0}$$

$$\underline{f'(x_0) \neq \text{any}}$$

$$f(x) = |x|$$

$$f'(0) = \underline{\text{any?}}$$



$$f(x) \geq 0$$

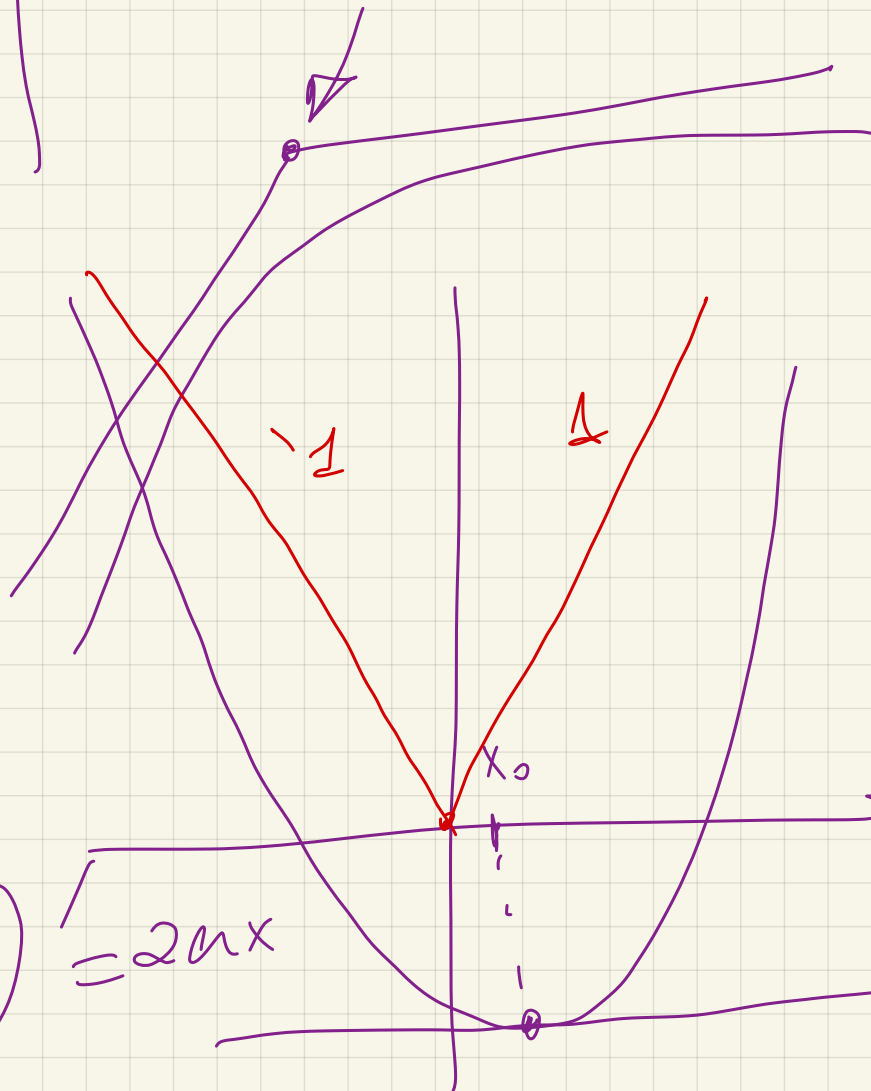
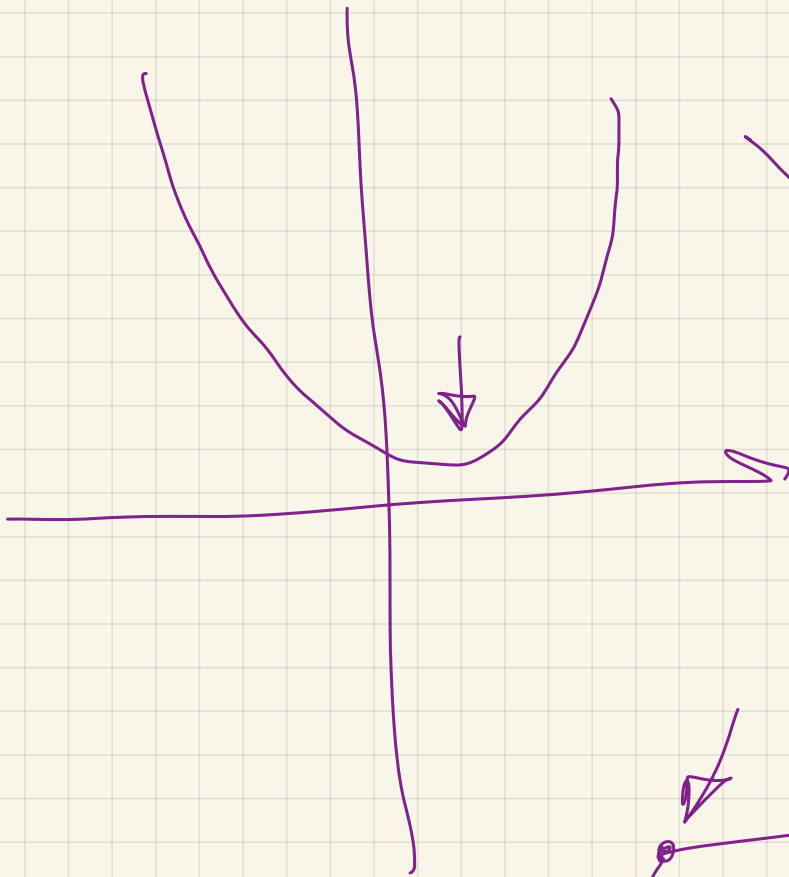
$$f(x) = x$$

$$\underline{f'(x) = 1}$$

$$f(x) \leq 0$$

$$f(x) = -x$$

$$f'(x) = \underline{\underline{-1}}$$



$$(a \cdot x^2)' = 2ax$$

$$f(x; y) = x^2 + y^2 + 2xy$$

$$\frac{\partial f}{\partial x} = 2x + 0 + 2y = 2x + 2y$$

$$z = -2x^3 + 2x^2 - 3y^2 - 6y$$

→ (1; -1) т. н. экстремум

$$\frac{\partial z}{\partial x} = -6x^2 + 4x$$

$$\frac{\partial z}{\partial y} = -6y - 6$$

Найдем значение экстр. ⇒

$$\frac{\partial z}{\partial x}(x) = 0$$

$$\text{и } \frac{\partial z}{\partial y}(x) = 0$$

б. x_0

$$\frac{\partial z}{\partial x}(1) = -6 \cdot 1 + 4 \cdot 1 = 0$$

$$\frac{\partial z}{\partial y}(-1) = -6(-1) - 6 = 6 - 6 = 0$$

Even M.Y. does $\rightarrow$ known
resp.

Do you know you. resp.

$$A = \frac{\partial^2 z}{\partial x^2}$$

$$B = \frac{\partial^2 z}{\partial x \partial y}$$

$$C = \frac{\partial^2 z}{\partial y^2}$$

Min: $D = AC - B^2 > 0$
 $A > 0$

Max: $D > 0$
 $A < 0$

$$\frac{\partial z}{\partial x} = -6x^2 + 6x$$

hyperbolic proper
hyperbolic
box

$$A = \frac{\partial^2 z}{\partial x^2} = \frac{\partial z}{\partial x} \left(\frac{\partial z}{\partial x} \right) = -12x + 6$$

$$C = \frac{\partial^2 z}{\partial y^2} = \frac{\partial z}{\partial y} \left(\frac{\partial z}{\partial y} \right) = -6$$

$$\frac{\partial z}{\partial y} = -6y - 6$$

$$B = \frac{\partial^2 z}{\partial x \partial y} = \frac{\partial}{\partial x} \left(\frac{\partial z}{\partial y} \right) =$$

$$(-6y - 6)^{1x} = 0$$

$$D = A^2 - B^2 = -6(-12x + 6)$$

$$= 72x - 36 \Rightarrow \underline{\underline{72 - 36 > 0}}$$

$$(1, -1)$$

$$D > 0$$

$$A = -12x + 6 = -12 + 6 = \underline{\underline{-6 < 0}}$$

$$f(x, y, \dots) \rightarrow \begin{cases} 1. \frac{\partial f}{\partial x} = 0 \\ \frac{\partial f}{\partial y} = 0 \end{cases}$$

$$\text{Eigen } g_a \rightarrow \begin{cases} A = \frac{\partial^2 f}{\partial x^2} \\ C = \frac{\partial^2 f}{\partial y^2} \\ B = \frac{\partial^2 f}{\partial x \partial y} \end{cases} \Rightarrow AC - B^2 = D$$

$$D > 0 \quad A > 0 \rightarrow \text{Min}$$

$$D > 0 \quad A < 0 \rightarrow \text{Max}$$

$$D < 0 \rightarrow \text{unstab. v. sel. Wert}$$